

SMART SEAPORT & LOGISTIC NETWORK INFRASTRUCTURE

Completed By

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LAB PROJECT REPORT

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Chapter - 01

Introduction

This chapter provides an overview of the DBMS project, including its objectives, scope, and significance. It sets the foundation by explaining the problem domain and the goals the project aims to achieve.

1.1 Introduction

Modern enterprise operations require secure and scalable networks to connect geographically dispersed locations. This project presents the design and Cisco Packet Tracer simulation of a multi-site network for a logistics firm, linking a central Headquarters (HQ) campus to two remote Branch offices. The HQ network features twelve departmental VLANs using Router-on-a-Stick inter-VLAN routing, connected via a Layer-3 Core Switch to an Edge Router running NAT Overload. The Branch offices implement localized VLANs for key departments and connect to HQ over serial WAN links.

Enterprise-wide routing is governed by a multi-area OSPF deployment, designating HQ as Backbone Area 0 and the remote branches as Areas 1 and 2 using Area Border Routers (ABRs). Centralized IP addressing is provided by a Data Center DHCP server leveraging `ip helper-address` relays across all sub-interfaces. This report documents the network topology, IP addressing scheme, configurations, and verification results.

1.2 Motivation

Single flat networks pose severe security risks, broadcast congestion, and operational bottlenecks for expanding enterprises. To address these issues, modern organizations require a segmented, hierarchical network architecture that enforces departmental traffic isolation, optimizes IP address space, centralizes core services like DHCP and Internet access, and provides fast fault convergence.

This project was undertaken to translate these enterprise requirements into a functional, production-style network using Cisco Packet Tracer. By integrating key networking principles—including VLAN segmentation, Router-on-a-Stick inter-VLAN routing, hierarchical Core-Distribution-Access topology, multi-area OSPF, NAT Overload, and DHCP relaying—this design demonstrates how individual network protocols operate within a unified, scalable, and secure enterprise infrastructure.

1.3 Objectives

The key objectives of this project are:

- **Design a Hierarchical Architecture:** Build a scalable Core–Distribution–Access network that easily accommodates future departments and branches with minimal redesign.
- **Ensure Traffic Isolation:** Implement VLAN segmentation across twelve HQ and eight branch departments using IEEE 802.1Q trunking and Router-on-a-Stick to contain broadcast domains and enforce security.
- **Deploy Multi-Area OSPF:** Enable dynamic routing for automatic link convergence and seamless multi-branch scalability.
- **Centralize Shared Services:** Streamline network administration by deploying a centralized Data Center DHCP server using ip helper-address relay across all sub-interfaces.
- **Configure Secure Internet Egress:** Implement NAT Overload (PAT) at the edge router to map private internal IP addresses to a public interface for controlled web access.
- **Establish WAN & Wireless Connectivity:** Connect HQ to two remote branches via dedicated serial WAN links and provide wireless coverage integrated into the wired LAN.
- **Incorporate Baseline Security:** Protect the internal network through VLAN segmentation, isolated Guest access, and restricted NAT translation ranges.

1.4 Feasibility Study

A review of academic and industry network designs confirms that hierarchical, VLAN-segmented, and OSPF-routed topologies are the standard for scalable enterprise networks. Cisco's enterprise campus architecture supports the Core-Distribution-Access model used here, where access switches handle end-device connectivity, distribution routers manage inter-VLAN routing, and the core layer aggregates WAN and Internet egress traffic. Similar implementation models validate that combining VLAN trunking, Router-on-a-Stick, multi-area OSPF, NAT Overload, and ip helper-address DHCP relays offers a cost-effective, highly verifiable solution within Cisco Packet Tracer. Consequently, this design was confirmed as technically sound and fully achievable within the scope of the Computer Networks Lab curriculum.

1.5 Gap Analysis

Table 1.1: Gap Analysis

Aspect	Literature / Standard Designs	Proposed Project
Scope	Single-building, isolated lab exercises.	Integrated multi-site enterprise (HQ + 2 Branches).
VLAN Segment	Small, localized VLANs.	12 HQ VLANs + 8 Branch VLANs with Router-on-a-Stick.
Routing	Single-area OSPF or static routing.	Multi-area OSPF (Area 0, Area 1, Area 2).
DHCP Management	Local, per-subnet DHCP servers.	Centralized Data Center DHCP via ip helper-address.
Edge Security	Basic, static NAT configurations.	Dynamic NAT Overload with scalable ACLs across sites.
Protocol Synergy	Features evaluated in isolation.	Full real-world interaction of OSPF, DHCP, and NAT.

1.6 Project Outcome

- **Simulated Multi-Site Network:** Delivered a fully configured, functional enterprise architecture in Cisco Packet Tracer linking HQ and two Branch offices.
- **VLAN Segmentation:** Achieved complete departmental traffic isolation using IEEE 802.1Q trunking and Router-on-a-Stick inter-VLAN routing.
- **Converged Dynamic Routing:** Implemented a multi-area OSPF domain spanning HQ Backbone Area 0 and Branch Areas 1 and 2.
- **End-to-End Connectivity:** Verified reliable inter-VLAN, cross-branch, and external internet reachability via Ping, Traceroute, and simulation tests.
- **Centralized Services & NAT:** Deployed centralized DHCP relay (ip helper-address) and NAT Overload for controlled public internet access across all internal subnets.
- **Integrated Wireless Access:** Configured a VLAN-aware wireless segment seamlessly bridged into the wired local network.
- **Technical Documentation:** Produced comprehensive technical documentation providing a baseline for future upgrades like IPv6, VPNs, and cloud integration.

Chapter - 02

Network Architecture

This chapter describes the overall system design and architecture of the enterprise network project. It outlines the major components — Headquarters, Branch 1, Branch 2, the Core, Distribution, and Access layers — their interactions, and the addressing and routing schemes used to build a scalable, secure, and efficient multi-site network.

2.1 Requirement Analysis & Design Specification

2.1.1 Requirements

To fulfill the core objectives, the network implements a three-tier Cisco hierarchical model supporting 12 HQ and 8 Branch departments:

- **Core Layer:** Centered around a Layer-3 HQ Core Switch that aggregates departmental and Edge routers over high-speed, routed point-to-point links.
- **Distribution Layer:** Comprises HQ routers (R2–R10) and Branch distribution routers running OSPF and executing Router-on-a-Stick inter-VLAN routing over 802.1Q trunks.
- **Access Layer:** Provides direct VLAN-tagged connectivity for end-user workstations, IP cameras, and wireless access points.

Site Topology:

- **Headquarters (HQ):** Manages 12 isolated VLANs—Administration, HR, Finance, Customs, Cargo Operations, Warehouse, Security, IT, Data Center, Logistics Planning, CCTV, and Guest.
- **Branches 1 & 2:** Replicate four core departments (Administration, HR, Finance, and Security) using local VLANs and connect to the HQ Edge Router via dedicated serial WAN links.

2.1.2 Network Topology Diagram

The enterprise architecture spans the ISP router, HQ Edge router, HQ Core Switch, nine HQ departmental routers with corresponding access switches, two Branch gateway routers with four distribution routers, and an access point bridged into the Finance **FIG: Network Topology Diagram**

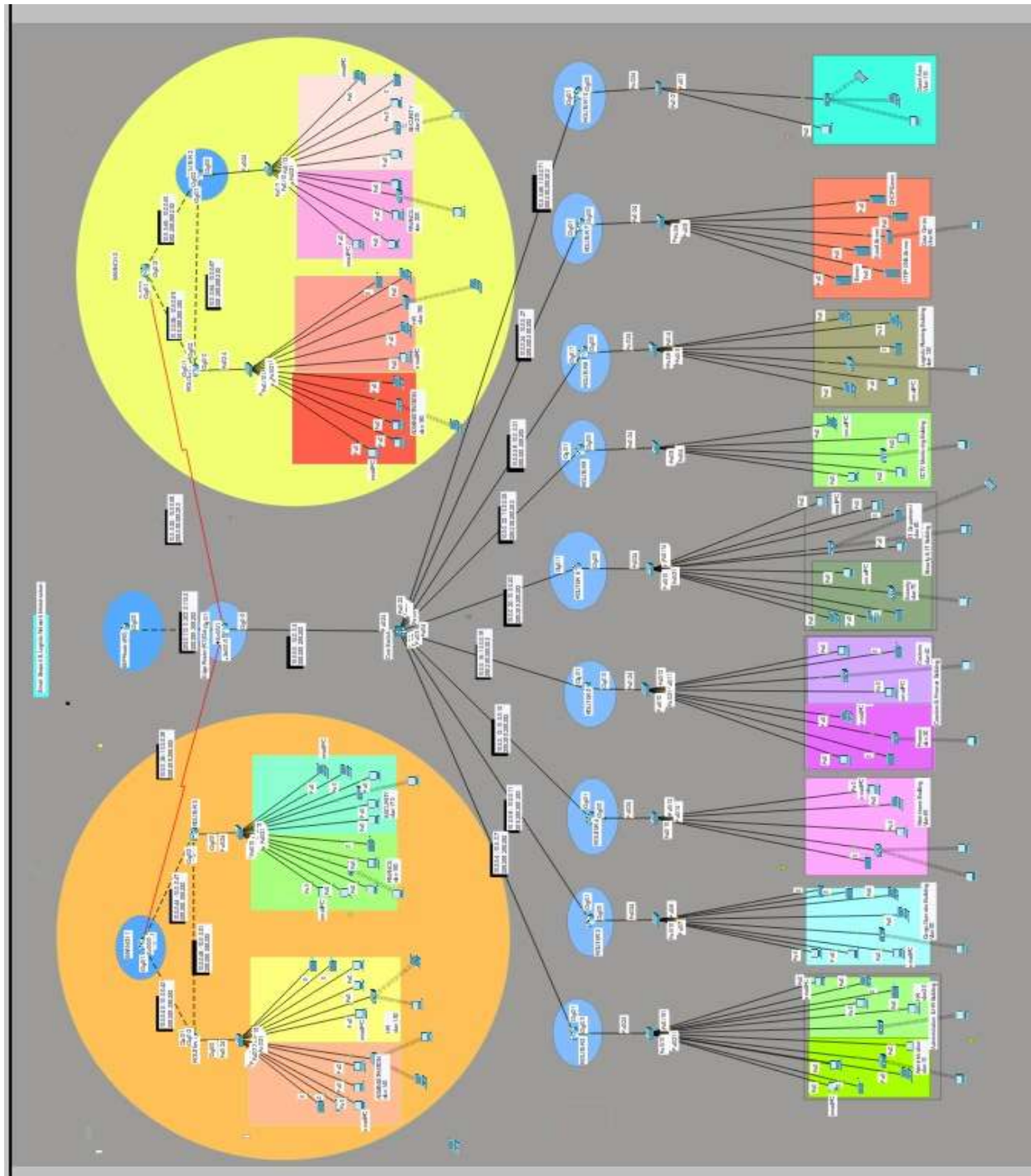


Figure: Complete Network Topology Diagram

Headquarters (HQ) Topology:

- **Access to Distribution:** Access switches connect to distribution routers via 802.1Q trunk links (Fa0/24), while end devices statically connect to dedicated VLAN access ports (Fa0/1–Fa0/20).
- **Inter-VLAN Routing:** Departmental routers employ Router-on-a-Stick sub-interfaces on Gig0/0 for VLAN routing, and use routed Gig0/1 interfaces to connect to the Core Switch.
- **Core Aggregation:** The Layer-3 Core Switch (ip routing) operates ten routed (no switchport) interfaces to aggregate the Edge router and nine departmental routers.

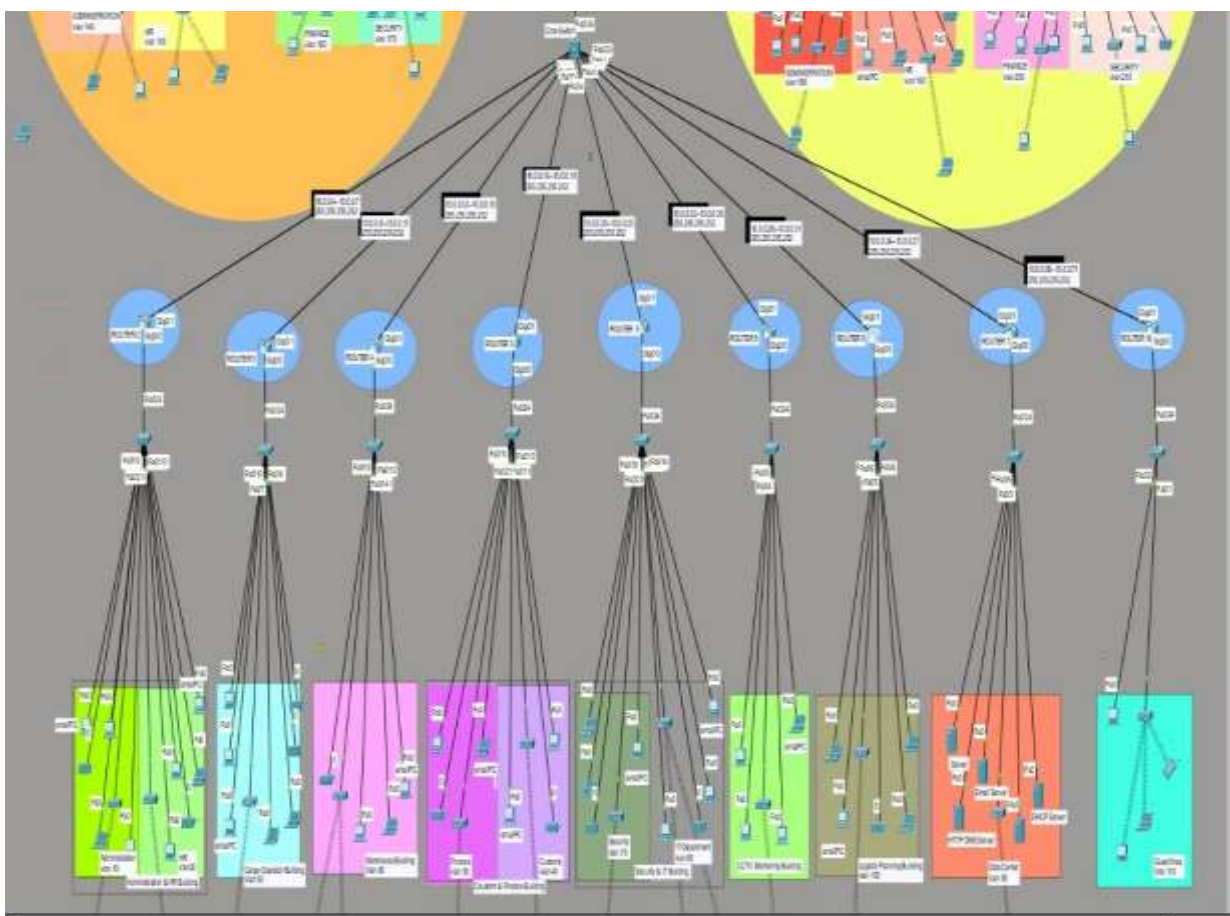


Figure: Headquarter Topology

Branch Topology:

- **WAN to Gateway:** Branch gateways (Branch1-Main / Branch2-Main) terminate serial WAN links from the HQ Edge router and connect to two internal distribution routers over Gigabit Ethernet.
- **Redundancy & Routing:** Intra-branch distribution routers are interconnected via dedicated Gigabit links (10.0.0.48/30 for Branch 1; 10.0.0.64/30 for Branch 2) to provide OSPF path redundancy.
- **Local Access:** Each branch distribution router manages two localized VLANs (Admin/HR or Finance/Security) using Router-on-a-Stick sub-interfaces and trunked access switches.

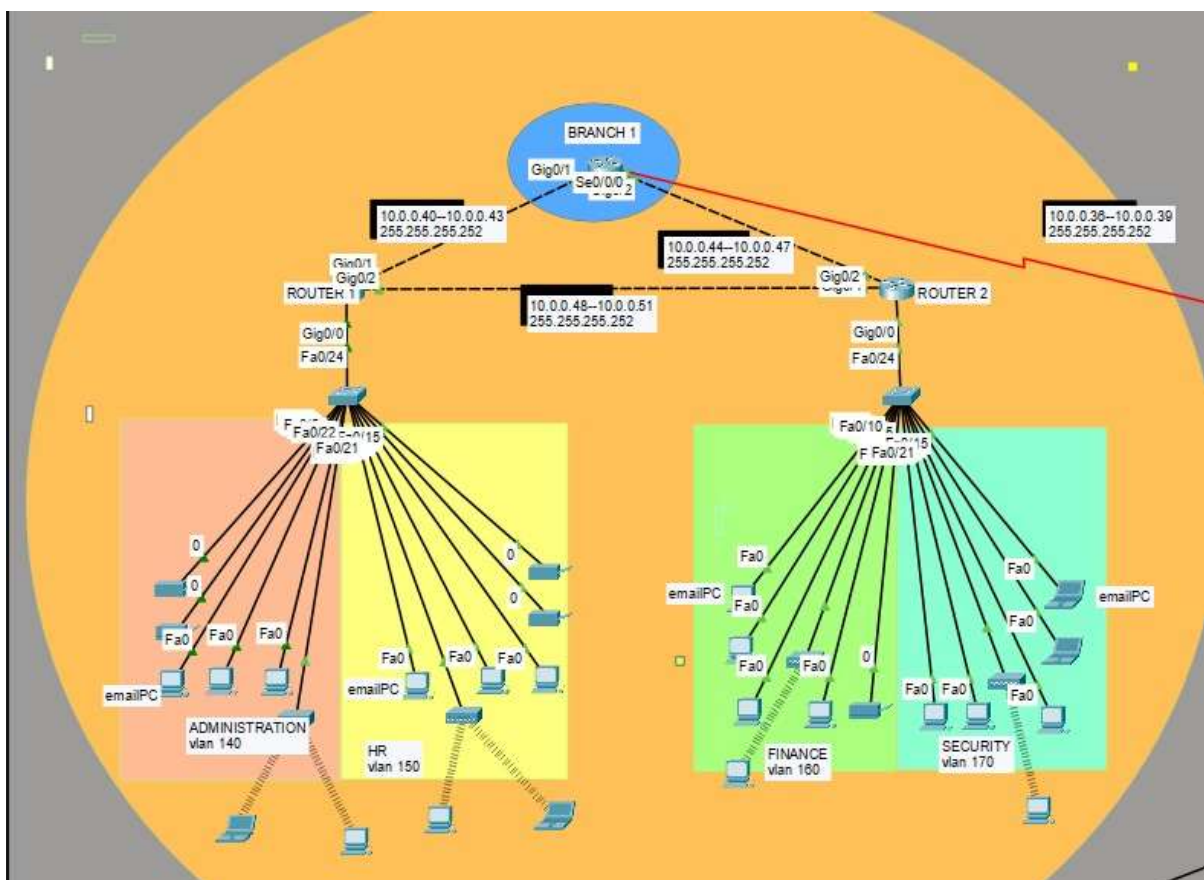


Figure: Branch-1 Topology

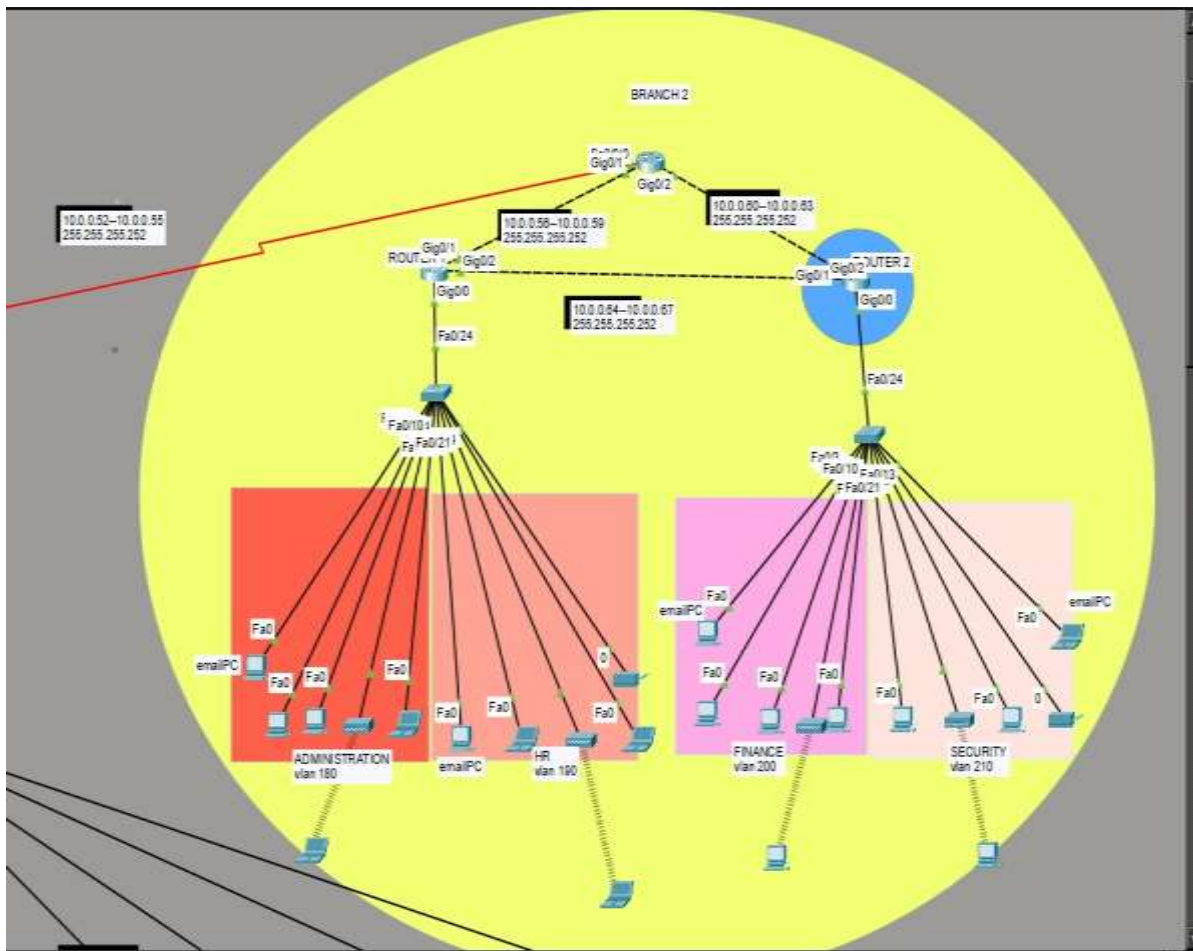


Figure: Branch-2 Topology

2.1.3 IP Addressing & Subnetting Table

The addressing architecture uses a Variable Length Subnet Masking (VLSM) strategy divided into two main private ranges and a public WAN pool:

- **Point-to-Point Links (10.0.0.0/24):** Subnetted into 19 consecutive /30 blocks (2 usable host IPs per link) for all router-to-router and router-to-switch interconnects to eliminate IP waste.
- **VLAN Segments (192.168.0.0/16):** Divided into /24 subnets per VLAN (accommodating up to 254 hosts per segment). For operational clarity, the 3rd octet matches the VLAN ID (e.g., VLAN 90 uses 192.168.90.0/24).
- **Public Egress (203.0.113.0/30):** A public /30 block assigned to the ISP-facing WAN link, serving as the pool for NAT Overload (PAT) translation on the Edge Router.

Table 2.1: Point-to-Point (WAN/Core) IP Addressing Table

Network / Mask	Link Description	Interface IPs (A / B)
10.0.0.0/30	Core-Switch Fa0/24 ↔ Edge-R1 Gig0/0	10.0.0.2 / 10.0.0.1
10.0.0.4/30	Core-Switch Fa0/1 ↔ R2-Admin-HR Gig0/1	10.0.0.5 / 10.0.0.6
10.0.0.8/30	Core-Switch Fa0/2 ↔ R3-Cargo Gig0/1	10.0.0.9 / 10.0.0.10
10.0.0.12/30	Core-Switch Fa0/3 ↔ R4-Warehouse Gig0/1	10.0.0.13 / 10.0.0.14
10.0.0.16/30	Core-Switch Fa0/4 ↔ R5-Customs-Finance Gig0/1	10.0.0.17 / 10.0.0.18
10.0.0.20/30	Core-Switch Fa0/5 ↔ R6-Security-IT Gig0/1	10.0.0.21 / 10.0.0.22
10.0.0.24/30	Core-Switch Fa0/6 ↔ R7-Data-Center Gig0/1	10.0.0.25 / 10.0.0.26
10.0.0.28/30	Core-Switch Fa0/7 ↔ R8-Logistics Gig0/1	10.0.0.29 / 10.0.0.30
10.0.0.32/30	Core-Switch Fa0/8 ↔ R9-CCTV Gig0/1	10.0.0.33 / 10.0.0.34
10.0.0.36/30	Edge-R1 Se0/0/0 ↔ Branch1-Main Se0/0/0	10.0.0.37 / 10.0.0.38
10.0.0.40/30	Branch1-Main Gig0/1 ↔ Branch1-Router1 Gig0/1	10.0.0.41 / 10.0.0.42
10.0.0.44/30	Branch1-Main Gig0/2 ↔ Branch1-Router2 Gig0/1	10.0.0.45 / 10.0.0.46
10.0.0.48/30	Branch1-Router1 Gig0/2 ↔ Branch1-Router2 Gig0/2	10.0.0.49 / 10.0.0.50
10.0.0.52/30	Edge-R1 Se0/0/1 ↔ Branch2-Main Se0/0/0	10.0.0.53 / 10.0.0.54
10.0.0.56/30	Branch2-Main Gig0/1 ↔ Branch2-Router1 Gig0/1	10.0.0.57 / 10.0.0.58
10.0.0.60/30	Branch2-Main Gig0/2 ↔ Branch2-Router2 Gig0/1	10.0.0.61 / 10.0.0.62
10.0.0.64/30	Branch2-Router1 Gig0/2 ↔ Branch2-Router2 Gig0/2	10.0.0.65 / 10.0.0.66
10.0.0.68/30	Core-Switch Fa0/23 ↔ R10-Guest-Wireless Gig0/1	10.0.0.69 / 10.0.0.70
203.0.113.0/30	Edge-R1 Gig0/1 ↔ ISP-R0 Gig0/0 (public)	203.0.113.2 / 203.0.113.1

Table 2.2: VLAN Table

VLAN ID	VLAN Name	Site	Subnet	Default Gateway	Serving Router
10	Administration	Headquarters	192.168.10.0/24	192.168.10.1	R2-Admin-HR
20	HR	Headquarters	192.168.20.0/24	192.168.20.1	R2-Admin-HR
30	Finance	Headquarters	192.168.30.0/24	192.168.30.1	R5-Customs-Finance
40	Customs	Headquarters	192.168.40.0/24	192.168.40.1	R5-Customs-Finance
50	Cargo Operations	Headquarters	192.168.50.0/24	192.168.50.1	R3-Cargo
60	Warehouse	Headquarters	192.168.60.0/24	192.168.60.1	R4-Warehouse
70	Security	Headquarters	192.168.70.0/24	192.168.70.1	R6-Security-IT
80	IT Department	Headquarters	192.168.80.0/24	192.168.80.1	R6-Security-IT
90	Data Center	Headquarters	192.168.90.0/24	192.168.90.1	R7-Data-Center
110	Guest	Headquarters	192.168.110.0/24	192.168.110.1	R10-Guest-Wireless
120	CCTV	Headquarters	192.168.120.0/24	192.168.120.1	R9-CCTV
130	Logistics Planning	Headquarters	192.168.130.0/24	192.168.130.1	R8-Logistics
140	Administration	Branch 1	192.168.140.0/24	192.168.140.1	Branch1-Router1
150	HR	Branch 1	192.168.150.0/24	192.168.150.1	Branch1-Router1
160	Finance	Branch 1	192.168.160.0/24	192.168.160.1	Branch1-Router2

170	Security	Branch 1	192.168.170.0/24	192.168.170.1	Branch1-Router2
180	Administration	Branch 2	192.168.180.0/24	192.168.180.1	Branch2-Router1
190	HR	Branch 2	192.168.190.0/24	192.168.190.1	Branch2-Router1
200	Finance	Branch 2	192.168.200.0/24	192.168.200.1	Branch2-Router2
210	Security	Branch 2	192.168.210.0/24	192.168.210.1	Branch2-Router2

Table 2.3: Device Summary Table

Device Category	Qty	Identifiers	Role / Notes
ISP Router	1	ISP-R0	Simulates the Internet service provider, advertises a default route toward the enterprise.
Edge Router	1	Edge-R1	Perimeter router; performs NAT overload and connects HQ, Branch 1, Branch 2, and the ISP.
HQ Core Multilayer Switch	1	Core-Switch	Layer-3 switch aggregating all HQ departmental routers via routed (no switchport) interfaces.
HQ Departmental Routers	9	R2–R10	Router-on-a-Stick gateways, one per department/pair of departments, terminating 802.1Q sub-interfaces.
HQ Access Switches	9	Admin-HR, Cargo, Warehouse, Customs-Finance, Security-IT, Data-Center, CCTV, Logistics, Guest	Layer-2 access switches providing VLAN access ports and an 802.1Q trunk uplink to the departmental router.

Branch Gateway Routers	2	Branch1-Main, Branch2-Main	Area Border Routers linking the WAN serial link (Area 0) to the branch-internal OSPF area.
Branch Distribution Routers	4	Branch1-Router1/2, Branch2-Router1/2	Router-on-a-Stick gateways for the branch VLAN pairs (Admin/HR and Finance/Security).
Branch Access Switches	4	Switch 1/2 per branch	Layer-2 access switches carrying the branch VLANs to end devices.
Wireless Access Point	1	AP-Finance	Provides WiFi connectivity bridged into the wired Finance VLAN (VLAN 30).
End-user / Server Devices	40+	PCs, Laptops, DHCP/Data-Center server, CCTV/IP-camera hosts	Representative end devices used to validate addressing, DHCP, and reachability in every VLAN.

Table 2.4: Routing (OSPF Area) Summary Table

OSPF Area	Participating Devices	Networks Advertised
Area 0 (Backbone)	Edge-R1, Core-Switch, R2–R10	HQ core links (10.0.0.0/30–10.0.0.68/30), all HQ VLAN subnets, ISP-facing static default route redistributed via default-information originate
Area 1	Branch1-Main, Branch1-Router1, Branch1-Router2	Branch 1 internal links (10.0.0.40/30, 10.0.0.44/30, 10.0.0.48/30) and VLAN subnets 192.168.140.0/24–192.168.170.0/24
Area 2	Branch2-Main, Branch2-Router1, Branch2-Router2	Branch 2 internal links (10.0.0.56/30, 10.0.0.60/30, 10.0.0.64/30) and VLAN subnets 192.168.180.0/24–192.168.210.0/24

Two key protocols finalize the Layer-3 architecture to deliver internet connectivity and automated host management:

- **NAT Overload (PAT):** Configured on the Edge Router (Edge-R1) using access-list 1 to permit internal source ranges (10.0.0.0/8 and 192.168.0.0/16). Interface Gig0/1 is designated as ip nat outside toward the ISP, while internal core (Gig0/0) and branch serial links are set as ip nat inside, translating all enterprise egress traffic to a single public IP.
- **Centralized DHCP Relay:** Managed by a dedicated DHCP Server (192.168.90.3) located in the Data Center VLAN (192.168.90.0/24). Every departmental and branch gateway sub-interface is configured with ip helper-address 192.168.90.3 to relay DHCPDISCOVER broadcasts across OSPF boundaries directly to the central server.

2.2 Use of Modern Simulation Tools

Cisco Packet Tracer was chosen as the simulation platform for its precise, vendor-native Cisco IOS CLI behavior across routers, multilayer switches, and access switches. It enables comprehensive modeling of wired devices, wireless clients, and IP peripherals within a single topology, bypassing physical hardware requirements. The platform's CLI, topology canvas, and diagnostic suite provided an efficient environment to design and validate this 32-device network (17 routers, 14 switches, 1 access point).

Verification Facilities:

- **Ping:** Validated Layer-3 reachability across local VLANs, inter-VLAN paths, HQ core interconnects, cross-branch links, and the simulated external ISP host.
- **Traceroute:** Verified hop-by-hop forwarding paths to ensure OSPF selected optimal routes and transited designated Area Border Routers (ABRs) across area boundaries.
- **Simulation Mode:** Inspected packet encapsulation at individual steps, confirming 802.1Q trunk tagging, Edge NAT translations, and OSPF PDU exchanges during convergence.
- **Routing Table Analysis (show ip route, show ip ospf neighbor):** Confirmed FULL OSPF neighbor adjacencies, proper propagation of inter-area routes (O IA), and enterprise-wide dissemination of the default route from the Edge Router.

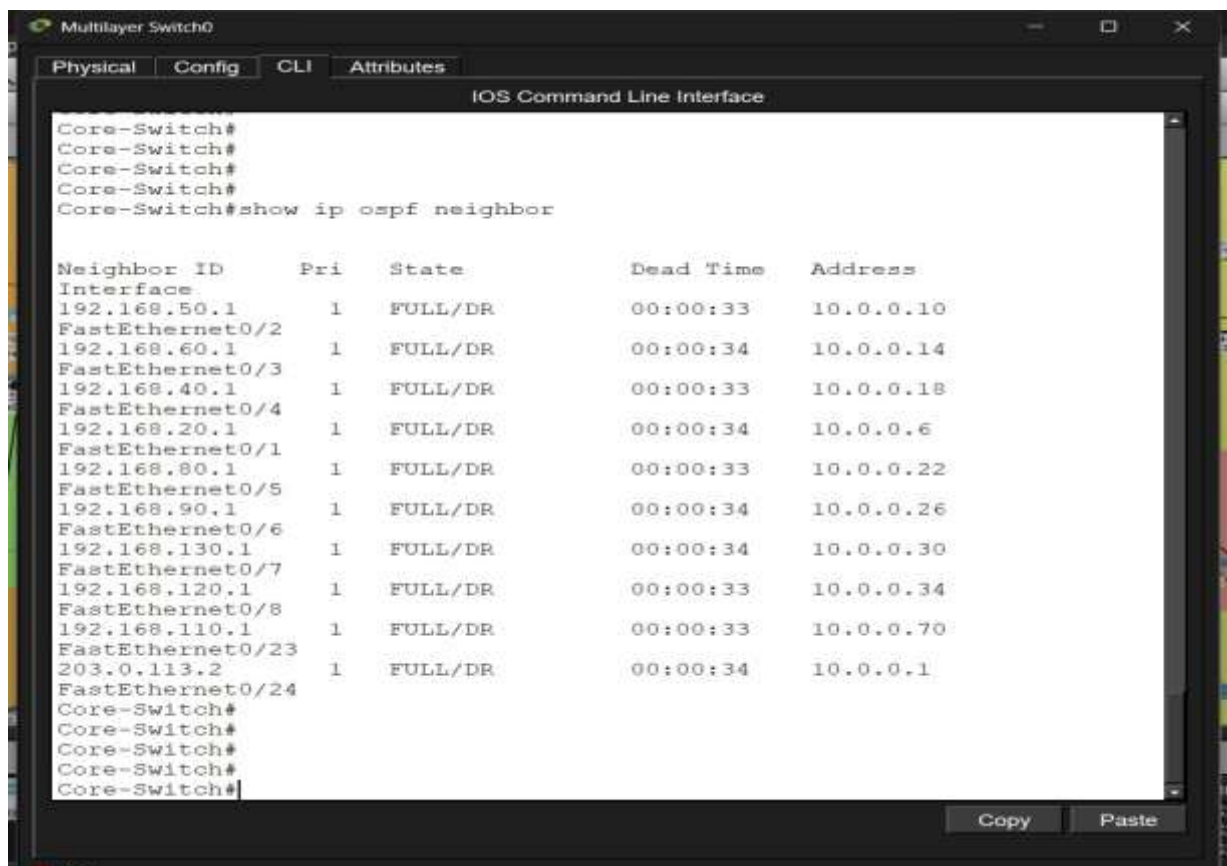


Figure: Ip OSPF Neighbor

Chapter - 03

Implementation and Results

This chapter details the step-by-step implementation process of the enterprise network project. It presents the configuration phases in the logical order they were carried out, the outputs obtained during verification, and a discussion of the resulting network's advantages, limitations, and performance characteristics.

3.1 Implementation

1. VLAN & Switch Port Assignment:

Created VLANs on access switches and assigned access ports to designated departmental VLANs.

2. 802.1Q Trunking:

Configured trunk ports (switchport mode trunk) on switch uplinks to carry multi-VLAN traffic.

3. Router-on-a-Stick (ROAS):

Created sub-interfaces on distribution routers (encapsulation dot1Q) to serve as default gateways per VLAN.

4. IP Addressing:

Assigned /30 addresses to point-to-point links and /24 gateway IPs to VLAN sub-interfaces.

5. Multi-Area OSPF:

Enabled router ospf 1, assigning HQ links to Area 0 and branch networks to Areas 1 and 2 via ABRs.

6. Edge NAT Overload:

Configured PAT on Edge-R1 matching ACL 1 internal ranges to translate egress traffic to a public IP.

7. Centralized DHCP Relay:

Applied ip helper-address 192.168.90.3 on all sub-interfaces to forward requests to the Data Center server.

8. Branch WAN Integration:

Linked branch gateways to Edge-R1 via serial links, verifying OSPF neighbor adjacencies across WANs.

9. Wireless Access Setup:

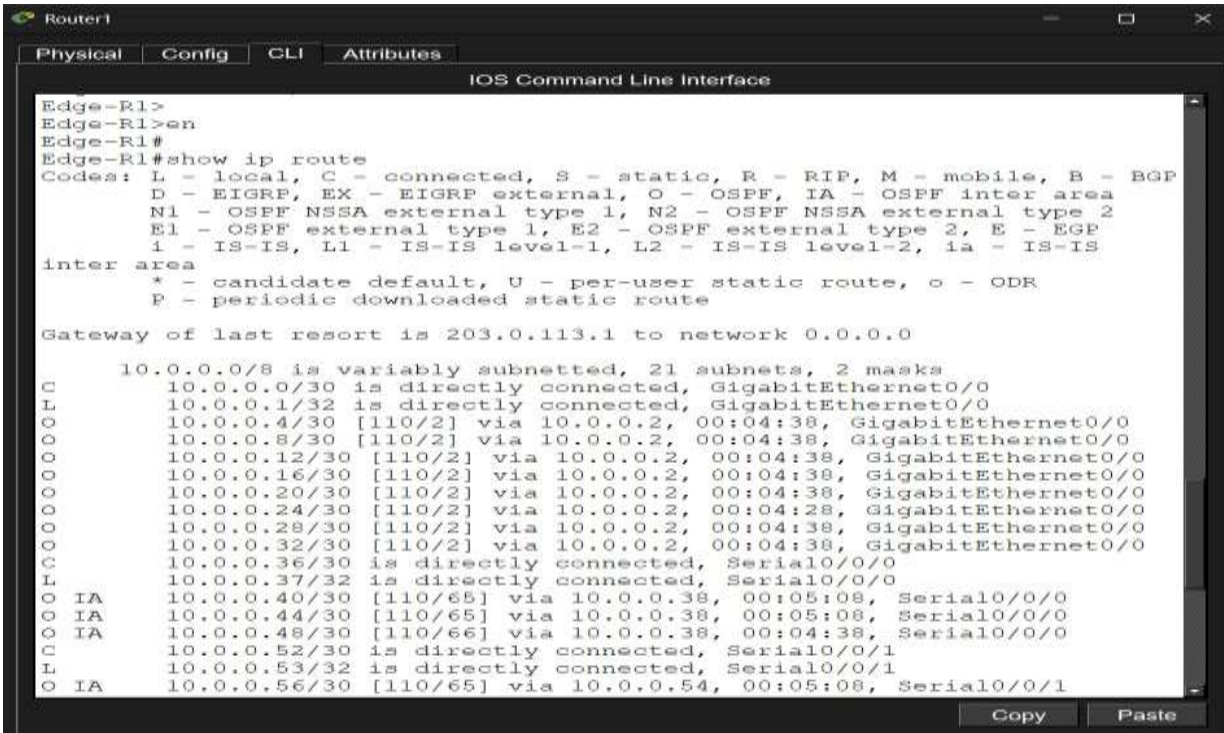
Connected a WAP to a VLAN 30 access port, set up SSID/security, and verified client IP auto-assignment.

10. End-to-End Verification:

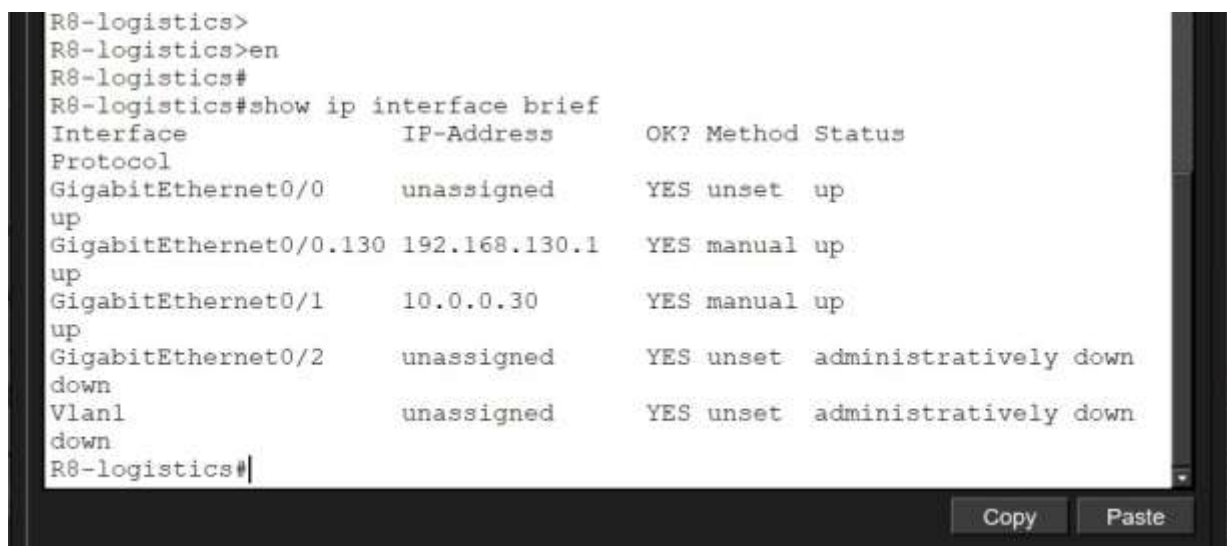
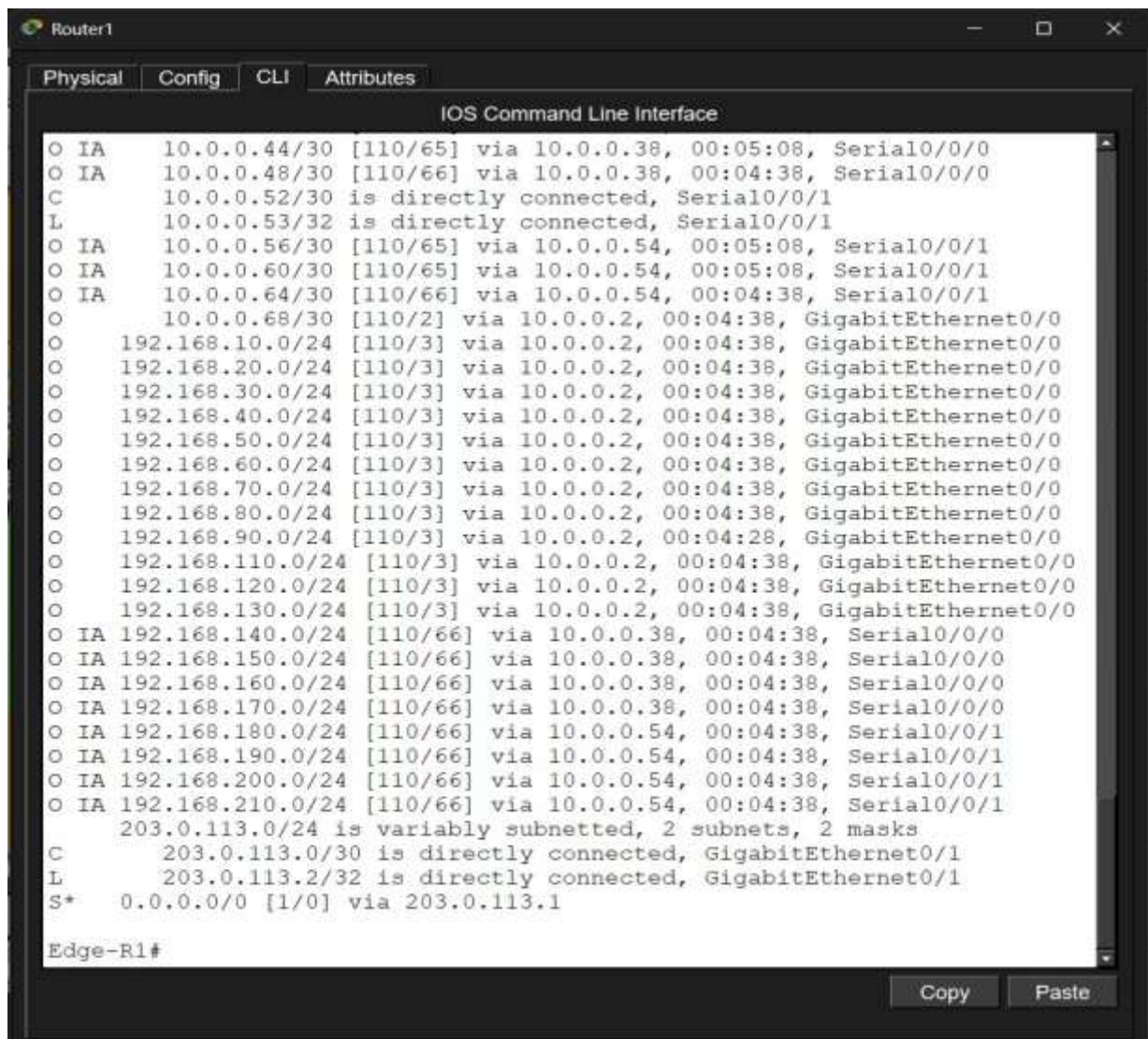
Performed Ping and Traceroute tests across local VLANs, cross-branch links, and the simulated ISP host.

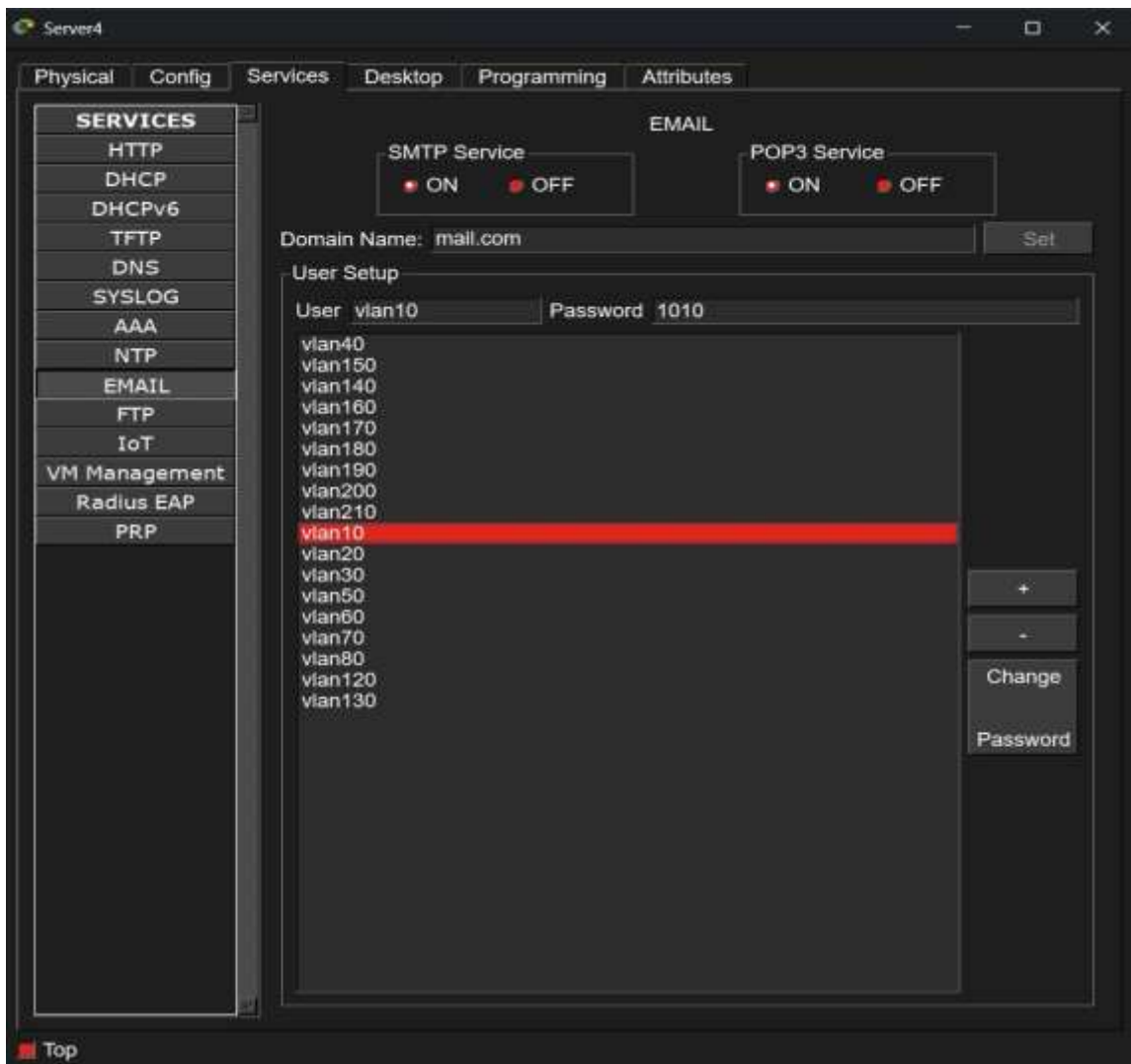
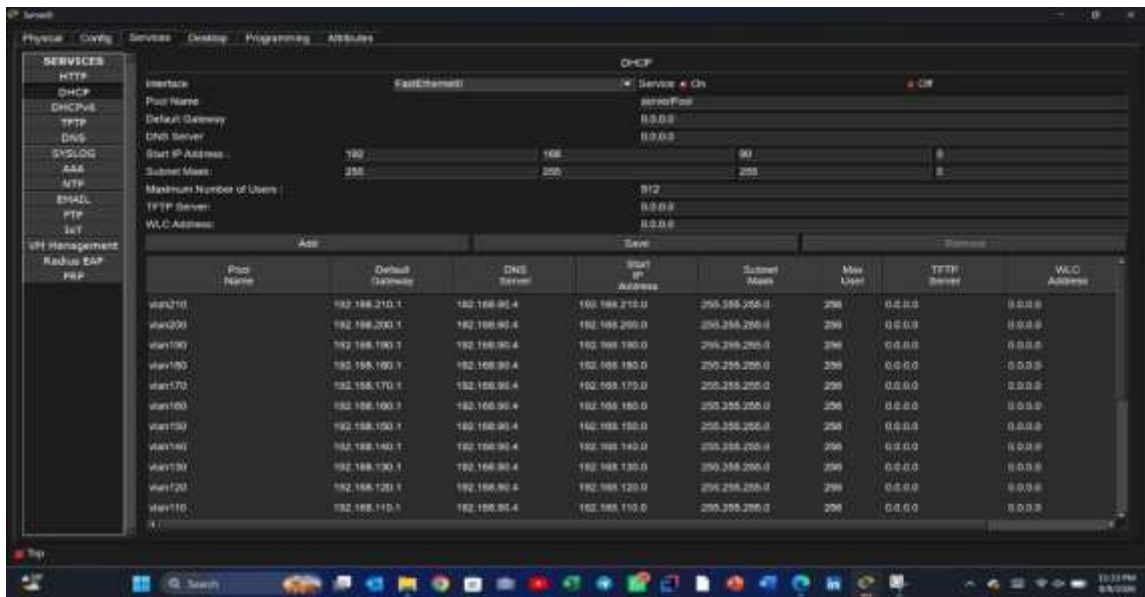


```
Switch0
Physical Config CLI Attributes
IOS Command Line Interface
Admin-HR-Switch>show vlan brief
VLAN Name Status Ports
-----
1 default active Fa0/22, Fa0/23, Gig0/1,
Gig0/2
10 Administration active Fa0/1, Fa0/2, Fa0/3,
Fa0/4
Fa0/5
Fa0/6
Fa0/7
Fa0/8
Fa0/9, Fa0/10
Fa0/11, Fa0/12, Fa0/13,
Fa0/14
Fa0/15, Fa0/16, Fa0/17,
Fa0/18
Fa0/19, Fa0/20, Fa0/21
1002 fddi-default active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active
Admin-HR-Switch>show interface trunk
Port Mode Encapsulation Status Native vlan
Fa0/24 on 802.1q trunking 1
Port Fa0/24
Vlans allowed on trunk
1-1005
Port Fa0/24
Vlans allowed and active in management domain
1,10,20
Port Fa0/24
Vlans in spanning tree forwarding state and not pruned
1,10,20
```



```
Router1
Physical Config CLI Attributes
IOS Command Line Interface
Edge-R1>
Edge-R1>en
Edge-R1#
Edge-R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS
inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is 203.0.113.1 to network 0.0.0.0
10.0.0.0/8 is variably subnetted, 21 subnets, 2 masks
C 10.0.0.0/30 is directly connected, GigabitEthernet0/0
L 10.0.0.1/32 is directly connected, GigabitEthernet0/0
O 10.0.0.4/30 [110/2] via 10.0.0.2, 00:04:38, GigabitEthernet0/0
O 10.0.0.8/30 [110/2] via 10.0.0.2, 00:04:38, GigabitEthernet0/0
O 10.0.0.12/30 [110/2] via 10.0.0.2, 00:04:38, GigabitEthernet0/0
O 10.0.0.16/30 [110/2] via 10.0.0.2, 00:04:38, GigabitEthernet0/0
O 10.0.0.20/30 [110/2] via 10.0.0.2, 00:04:38, GigabitEthernet0/0
O 10.0.0.24/30 [110/2] via 10.0.0.2, 00:04:28, GigabitEthernet0/0
O 10.0.0.28/30 [110/2] via 10.0.0.2, 00:04:38, GigabitEthernet0/0
O 10.0.0.32/30 [110/2] via 10.0.0.2, 00:04:38, GigabitEthernet0/0
C 10.0.0.36/30 is directly connected, Serial0/0/0
L 10.0.0.37/32 is directly connected, Serial0/0/0
O IA 10.0.0.40/30 [110/65] via 10.0.0.38, 00:05:08, Serial0/0/0
O IA 10.0.0.44/30 [110/65] via 10.0.0.38, 00:05:08, Serial0/0/0
O IA 10.0.0.48/30 [110/66] via 10.0.0.38, 00:04:38, Serial0/0/0
C 10.0.0.52/30 is directly connected, Serial0/0/1
L 10.0.0.53/32 is directly connected, Serial0/0/1
O IA 10.0.0.56/30 [110/65] via 10.0.0.54, 00:05:08, Serial0/0/1
```



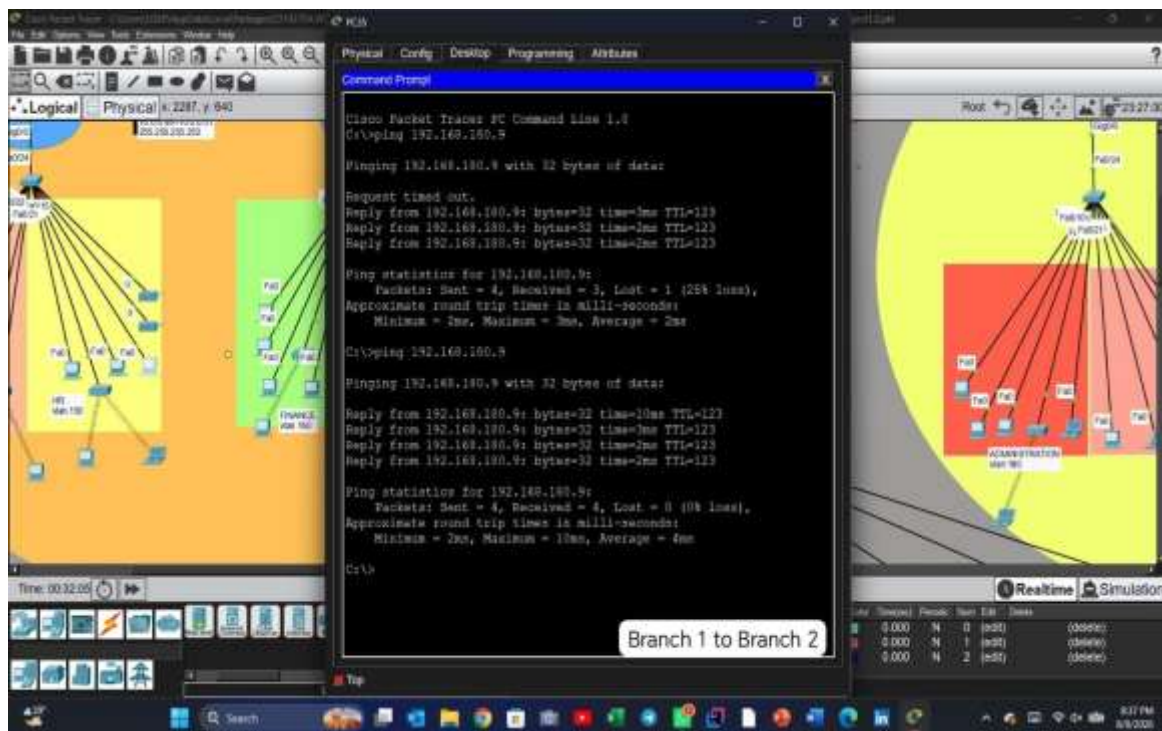


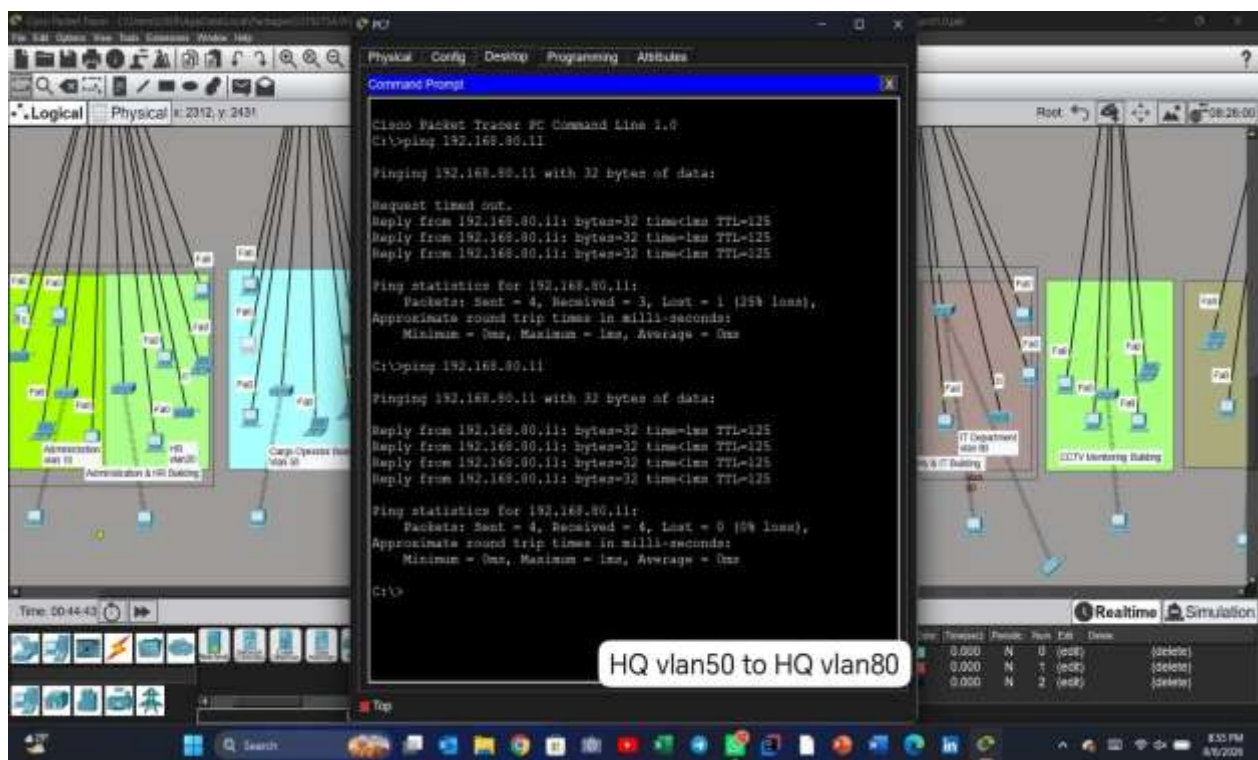
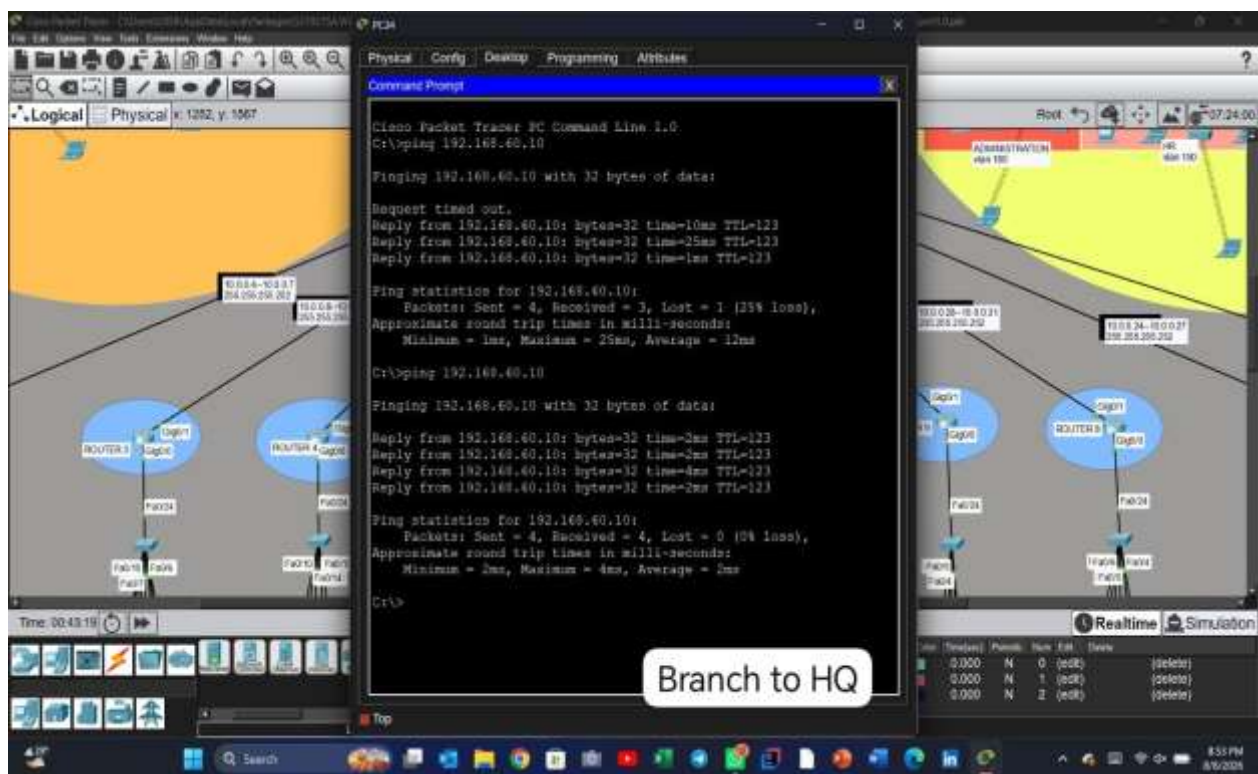
3.2 Output

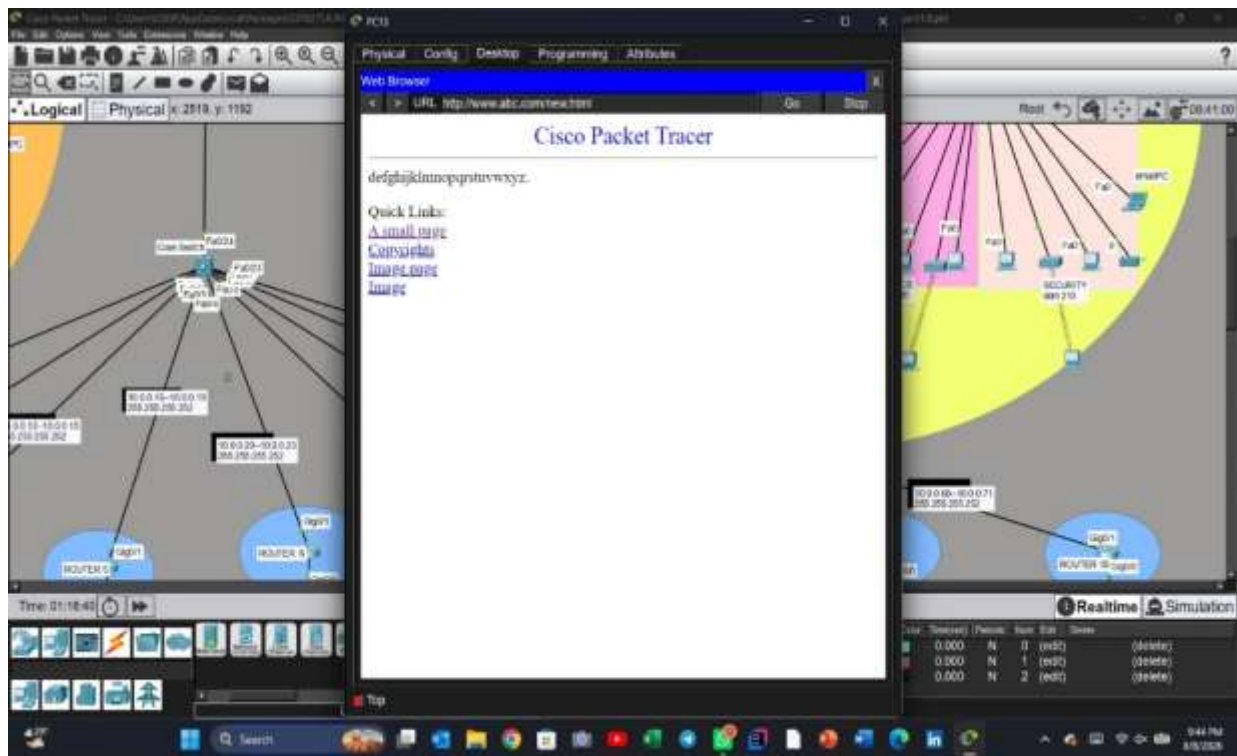
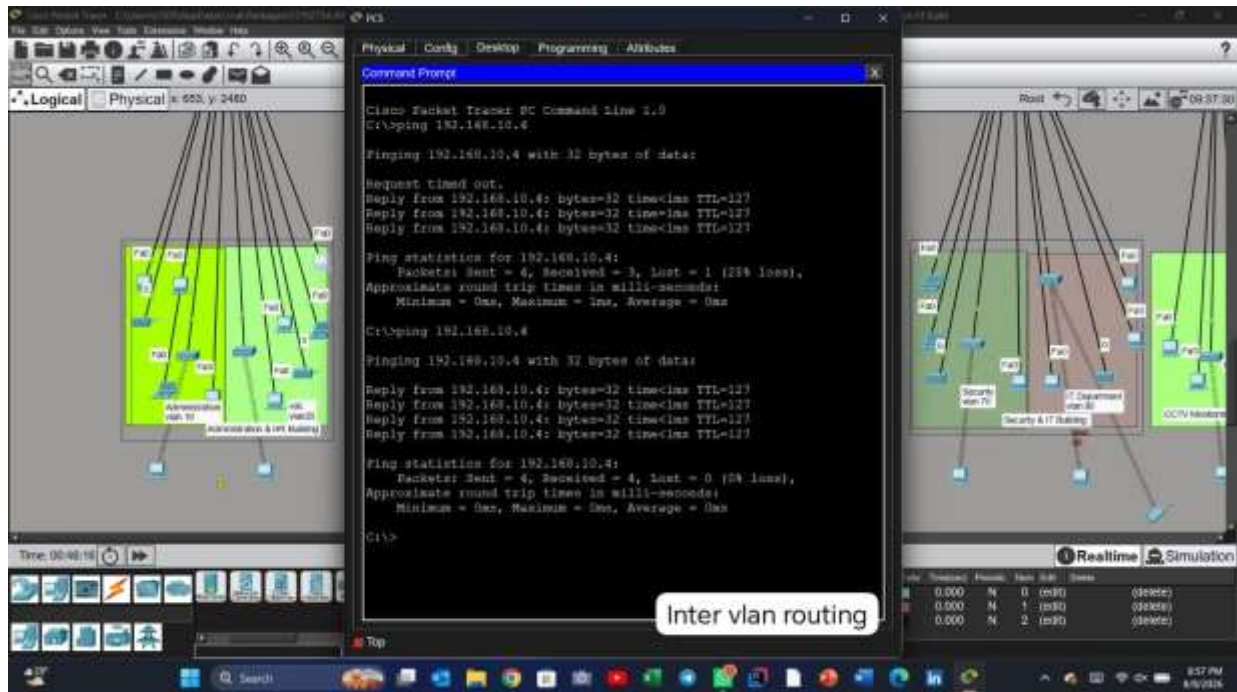
- **Ping Reachability:** Achieved 0% packet loss across intra-VLAN, inter-VLAN, cross-branch (HQ–Branch 1–Branch 2), and simulated external Internet paths post-OSPF convergence.
- **OSPF Adjacency (`show ip ospf neighbor`):** Confirmed `FULL` neighbor states enterprise-wide, with branch gateways maintaining dual `FULL` adjacencies toward Area 0 (HQ Edge) and Area 1/2 (internal distribution).
- **Internet Egress & PAT:** Verified all internal subnets successfully reached external Internet hosts beyond `ISP-R0`, translating internal IPs to public address `203.0.113.2`.
- **NAT Translations (`show ip nat translations`):** Confirmed active inside-local to inside-global mappings on `Edge-R1`, validating proper port-based translation (PAT) for active flows.
- **Path Forwarding (`traceroute`):** Confirmed inter-area traffic from Branch 1 to Branch 2 correctly transited `Branch1-Router → Branch1-Main → Edge-R1 → Branch2-Main → Branch2-Router`.
- **Wireless Client Validation:** Confirmed a wireless client associated with the SSID received an IP in `192.168.30.0/24` via central DHCP relay, successfully reaching local Finance hosts and the Internet.

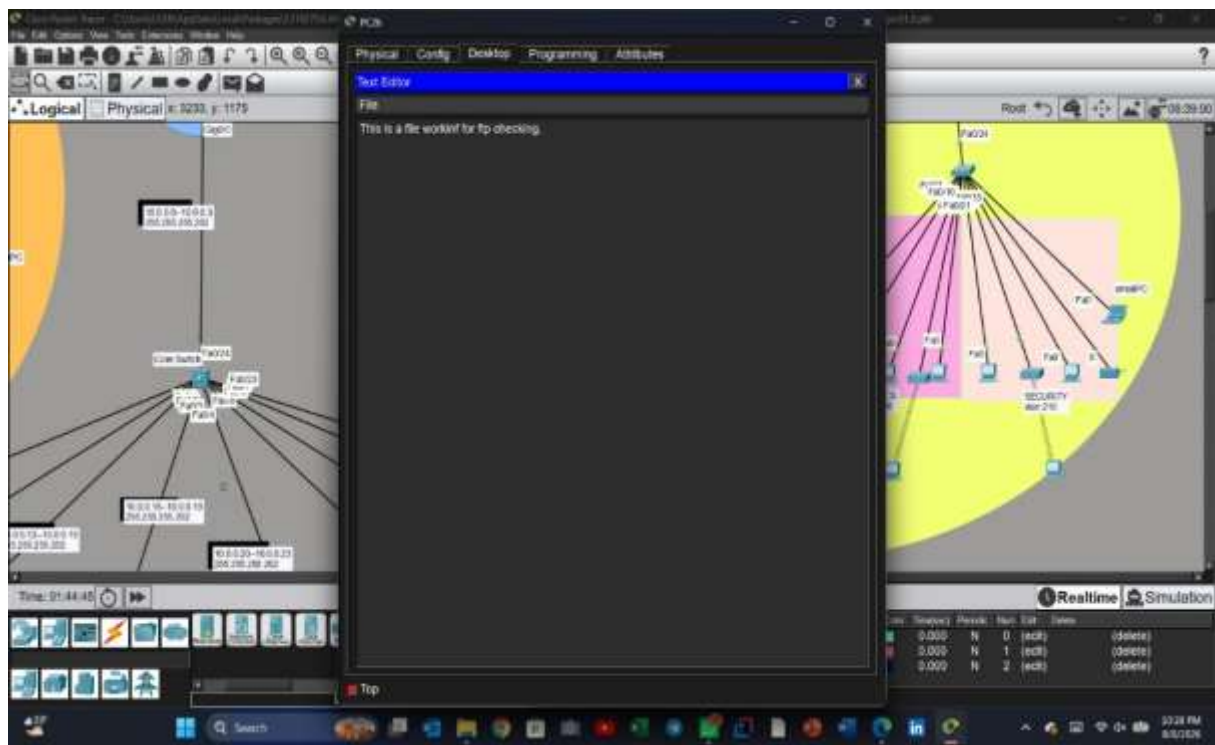
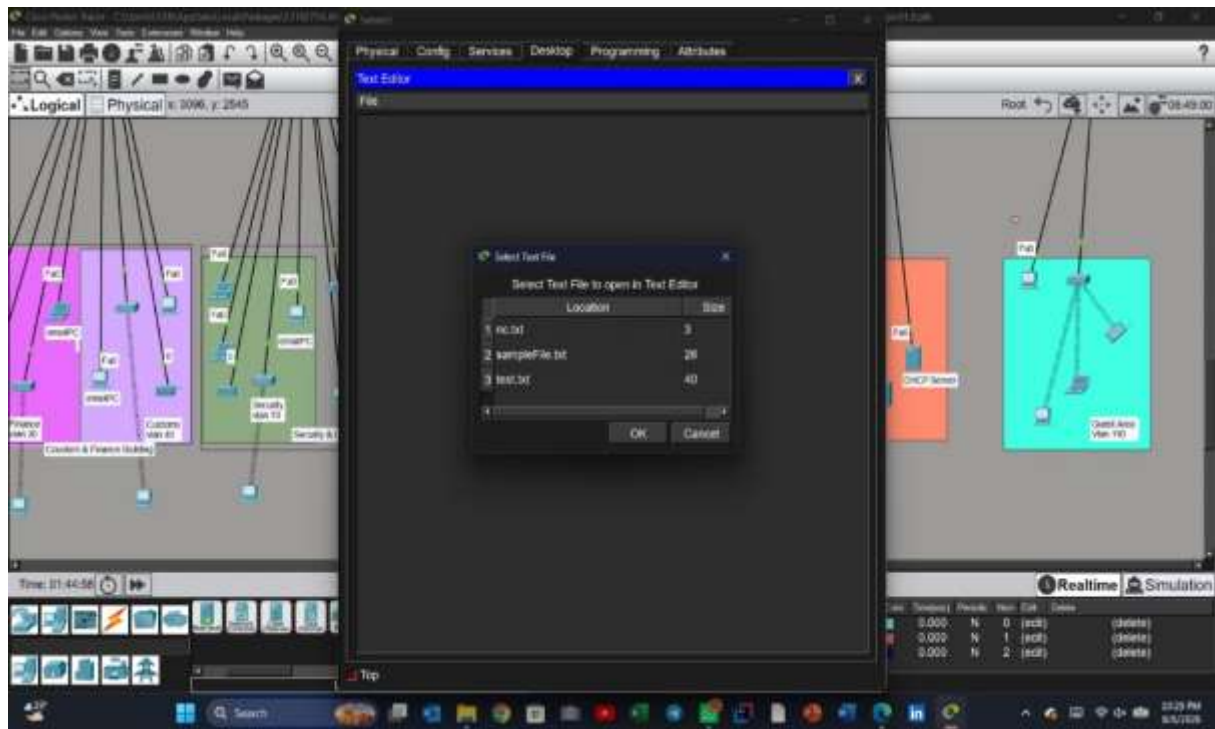
Table 3.1: Representative NAT Translation Table (Edge-R1)

Protocol	Inside Local	Inside Global	Outside Local	Outside Global
icmp	192.168.10.10:512	203.0.113.2:512	8.8.8.8:512	8.8.8.8:512
tcp	192.168.140.15:1500	203.0.113.2:1500	8.8.8.8:80	8.8.8.8:80
icmp	192.168.90.20:770	203.0.113.2:770	8.8.8.8:770	8.8.8.8:770









3.3 Discussion

This project successfully validates a scalable, multi-site enterprise network architecture in Cisco Packet Tracer by integrating a three-tier Core-Distribution-Access hierarchy across a 12-VLAN Headquarters campus and two 4-VLAN remote branches. By implementing departmental VLAN segmentation with 802.1Q trunking, Router-on-a-Stick inter-VLAN routing, and multi-area OSPF (Area 0 core, Areas 1 and 2 branches), the design delivers strict traffic isolation, rapid link convergence, and seamless multi-branch scalability. Operational efficiency is achieved through centralized Data Center DHCP relay (ip helper-address) and Edge NAT Overload for secure, controlled Internet egress. While the single Core Switch, Edge router, and DHCP server introduce single points of failure that require future hardware redundancy and firewall integration, performance testing confirmed 0% packet loss, valid PAT address translation, and optimal inter-area path selection, proving the technical viability of a unified enterprise network design.

Chapter - 04

Conclusion

4.1 Summary

This project successfully designed, implemented, and verified a scalable multi-site enterprise network using Cisco Packet Tracer to connect a 12-department Headquarters campus and two 4-department Branch offices. The architecture established 20 isolated VLANs using IEEE 802.1Q trunking and Router-on-a-Stick inter-VLAN routing, aggregated through a Layer-3 Core Switch at HQ. Enterprise-wide reachability was achieved via a multi-area OSPF routing domain (Area 0 core; Areas 1 & 2 branches). Essential network services were centralized, leveraging Edge NAT Overload for shared Internet egress and ip helper-address relays to forward DHCP requests from all subnets to a central Data Center server. Connectivity, wireless access, dynamic routing, and PAT translations were fully validated through Ping, Traceroute, Simulation mode, and routing table analysis, confirming all primary design objectives were met.

4.2 Limitation

- **Single Points of Failure:** Single Core Switch, Edge router, and DHCP server leave the entire enterprise vulnerable to outages.
- **NAT Capacity:** Single public IP for NAT Overload restricts outbound scalability and session tracking under heavy traffic.
- **Security Gaps:** Lack of firewalls, VPNs, and IDS/IPS means inter-departmental and WAN traffic relies entirely on basic VLAN/NAT boundaries.
- **IPv4 Only:** Network lacks dual-stack IPv6 implementation.
- **Configuration Overhead:** Overly broad OSPF network statements on the Core Switch require cleanup for production readiness.
- **Non-Redundant WAN:** Single serial WAN links per branch create isolated points of failure during link drops.

4.3 Future Work

Redundancy & High Availability: Deploy secondary Core/Edge devices, redundant WAN links, and HSRP/VRRP protocols.

Advanced Security: Implement dedicated zone-based firewalls, IPsec VPN tunnels, and IDS/IPS across critical VLANs.

IPv6 & BGP Integration: Add dual-stack IPv6 addressing and BGP multi-homing for multi-ISP resilience.

Cloud & Server Load Balancing: Establish cloud VPC connectivity and load balance Data Center services like DHCP.

Automation & Monitoring: Introduce Ansible/Cisco DNA automation alongside SNMP/NetFlow monitoring tools (LibreNMS/SolarWinds).

References

- [1] Cisco Systems. Cisco Networking Academy — CCNA: Switching, Routing, and Wireless Essentials. Cisco Press.
- [2] Cisco Systems. Enterprise Campus Infrastructure: Hierarchical (Core-Distribution-Access) Design Best Practices. Cisco Design Guides.
- [3] Cisco Systems. OSPF Design Guide — Multi-Area OSPF Network Design. Cisco Technical Documentation.
- [4] Cisco Systems. Configuring Network Address Translation (NAT) — Cisco IOS NAT Configuration Guide.
- [5] Cisco Systems. Cisco Packet Tracer User Guide, Cisco Networking Academy.
- [6] Jon Kleinberg and Eva Tardos. Algorithm Design. Pearson Education India, 2006.